

COLLEGE OF SCIENCE

PHYS2020

Simulation Plan Semester One 2026

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0.1 What I've done so far

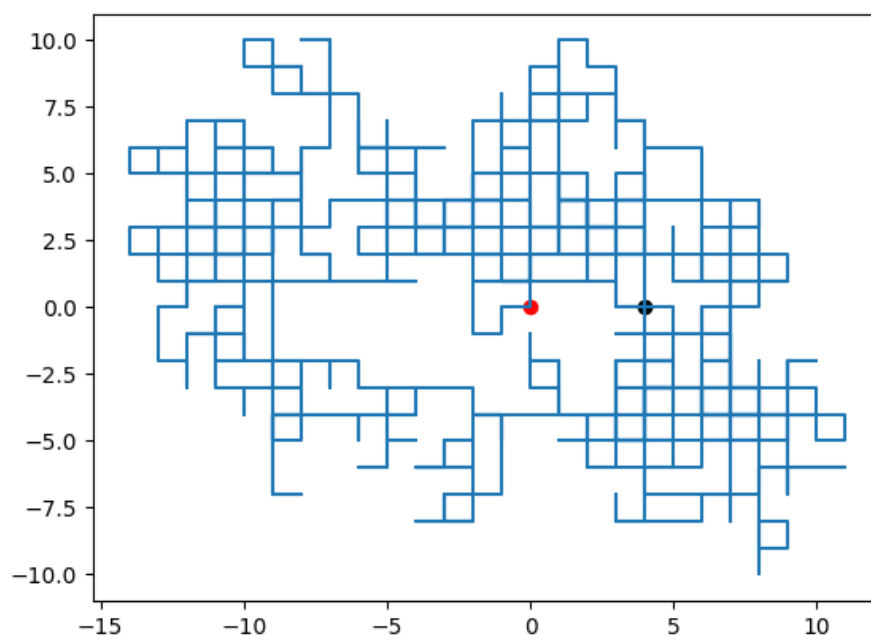
I have now set up a github repository which is also linked to my google drive
<https://github.com/carmel3141/Brownian-Motion-Simulation-PHYS2020>

I have tested a few commits to see if i remember how to use github.

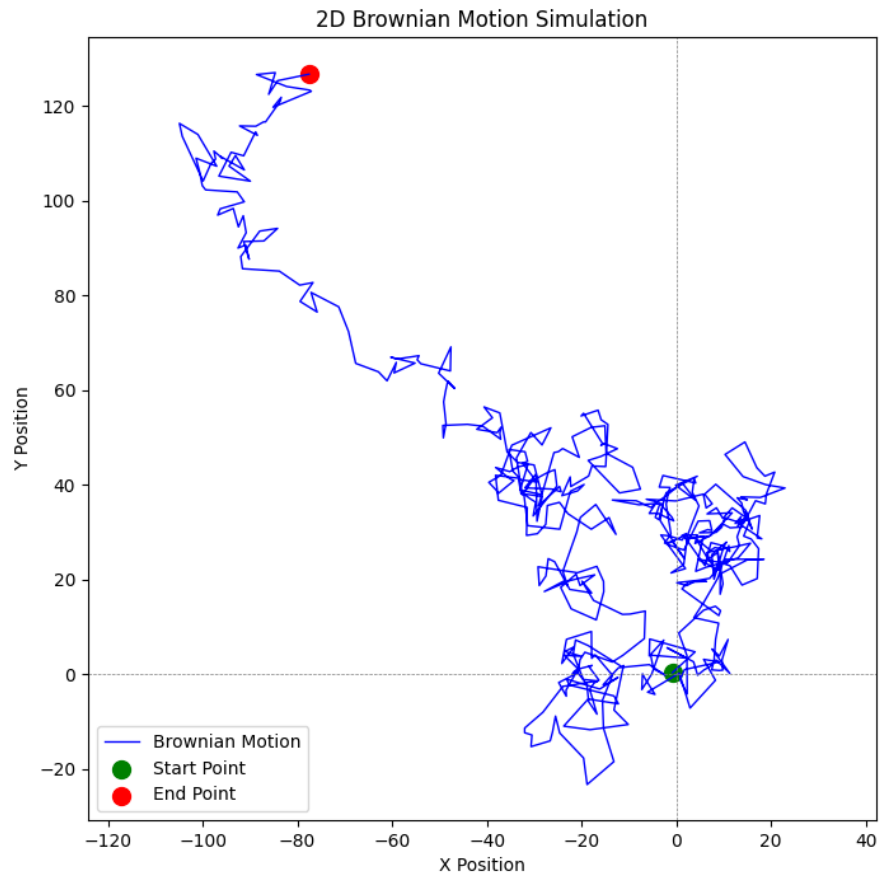
yeah so i accidentally bricked my git at one point but its fixed now

moved away from the lattice model to a continuous model, which is more realistic and will be more interesting to analyse after reading a couple of websites <https://ipython-books.github.io/133-simulating-a-brownian-motion/>
<https://www.quantstart.com/articles/brownian-motion-simulation-with-python/>

one of these has a continuous one where it uses a polar co-ordinate system to generate the random walk which would be interesting to try too



trial 1. added start point and end point, current version has legend too



now uses interpolation instead of grid/lattice so it's more continuous with time-steps, much better !!!